

AUTOMATED INTELLIGENT EMAIL SYSTEM

DATA ACCESS LAYER AND TESTING

PART A: DATA ACCESS LAYER (DAL) IMPLEMENTATION

1. Introduction

The Data Access Layer (DAL) acts as an interface between application and database. It manages storing and retrieving email data.

2. Database Design

Database: email_system

Table: emails (id, sender, content, cleaned_text, category, priority, response)

3. DAL Implementation

- Database connection using mysql.connector
- classify_email() for classification
- save_email() to store emails
- get_emails_by_category() to retrieve emails

4. Working

- Input email
- Classify
- Store in DB
- Retrieve

5. Advantages

- Separation of concerns
- Maintainability
- Security

PART B: TESTING

1. White Box Testing

Tested classify_email() logic.

Test Cases:

- "win lottery now" → Spam
- "I have a complaint" → Complaint
- "meeting tomorrow" → Important

Result: Passed

2. Black Box Testing

Tested full system behavior.

Test Cases:

- "You win lottery" → Spam
- "I have a complaint" → Complaint
- "Project meeting" → Important

Result: Passed

Final Conclusion:

System works correctly with proper classification and storage.